Marmara University
Engineering Faculty
Electrical and Electronics Engineering
EE4077: Fundamentals of Machine Learning
Midterm Exam 1 Solutions (Spring, 2024)

1.

(a) $P[Y=y \mid X_1=x_1, X_2=x_2, X_3=x_3] = P[Y=y] \cdot P[X_1=x_1 \mid Y=y] \cdot P[X_2=x_2 \mid Y=y] \cdot P[X_3=x_3 \mid Y=y]$ (4)

$P[Y=0 \mid X_1=0, X_2=1, X_3=1] = P[Y=0] \cdot P[X_1=0 \mid Y=0] \cdot P[X_2=1 \mid Y=0] \cdot P[X_3=1 \mid Y=0]$ (3)

$= \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{1}{27}$ (3)

(b) Laplace smoothing with $k=1$ gives

X_1	X_2	X_3	Y	
1	0	1	0	} given
0	0	1	0	
1	1	0	0	
0	1	0	0	} added
0	1	1	0	
0	0	0	1	} given
1	1	1	1	
1	0	0	1	
0	1	0	1	} added
0	1	1	1	

$P[Y=1 \mid X_1=0, X_2=1, X_3=0]$ (5)

$= P[Y=1] \cdot P[X_1=0 \mid Y=1] \cdot P[X_2=1 \mid Y=1] \cdot P[X_3=0 \mid Y=1]$

$= \frac{1}{2} \cdot \frac{3}{5} \cdot \frac{3}{5} \cdot \frac{3}{5}$ (5)

$= \frac{27}{250}$

2.

$$p(\mathcal{D}|\theta) = \frac{e^{-\theta}\theta^3}{3!}\frac{e^{-\theta}\theta^4}{4!}\frac{e^{-\theta}\theta^1}{1!}$$

$$\begin{aligned}
\hat{\theta}_{MAP} &= \arg\min_\theta -\log\left(p(\mathcal{D}|\theta)p(\theta)\right) \\
&= \arg\min_\theta -\log\left(\frac{e^{-\theta}\theta^3}{3!}\frac{e^{-\theta}\theta^4}{4!}\frac{e^{-\theta}\theta^1}{1!}\frac{1}{\Gamma(2)2^2}\theta^{(2-1)}e^{\frac{-\theta}{2}}\right) \\
&= \arg\min_\theta -\log e^{-3\theta}\theta^8\theta^{(2-1)}e^{\frac{-\theta}{2}} \\
&= \arg\min_\theta -\log e^{-3\theta-\frac{\theta}{2}}\theta^{8+\alpha-1} \\
&= \arg\min_\theta -\left((-3\theta-\frac{\theta}{2})+(8+2-1)\log(\theta)\right)
\end{aligned}$$

Setting the derivative equal to zero yields

$$0 = -3 - \frac{1}{2} + \frac{(7+2)}{\theta}$$

$$\implies \theta_{MAP} = \frac{7+2}{3+\frac{1}{2}} = 2.57142857143$$

3.

(a)

$H(Y) = -\frac{6}{8} * \log_2(\frac{6}{8}) - \frac{2}{8} * \log_2(\frac{2}{8}) \approx 0.811$

For attribute X_1,

- $H(Y \mid X_1 = sunny) = -[\frac{1}{3} * \log_2(\frac{1}{3}) + \frac{2}{3} * \log_2(\frac{2}{3})] \approx 0.918$
- $H(Y \mid X_1 = rain) = 0$
- $H(Y \mid X_1 = overcast) = 0$

$\implies H(Y \mid X_1) = [\frac{3}{8} * 0.918 + \frac{3}{8} * 0 + \frac{2}{8} * 0] \approx 0.344$
$\implies I(Y; X_1) \approx 0.811 - 0.344 = 0.467$

For attribute X_2,

- $H(Y \mid X_2 = hot) = -[\frac{1}{3} * \log_2(\frac{1}{3}) + \frac{2}{3} * \log_2(\frac{2}{3})] \approx 0.918$
- $H(Y \mid X_2 = cool) = 0$
- $H(Y \mid X_2 = mild) = -[\frac{3}{4} * \log_2(\frac{3}{4}) + \frac{1}{4} * \log_2(\frac{1}{4})] \approx 0.811$

$\implies H(Y \mid X_2) = [\frac{3}{8} * 0.918 + \frac{1}{8} * 0 + \frac{4}{8} * 0.811] \approx 0.75$
$\implies I(Y; X_2) \approx 0.811 - 0.75 = 0.061$

For attribute X_3,

- $H(Y \mid X_3 = high) = -[\frac{1}{2} * \log_2(\frac{1}{2}) + \frac{1}{2} * \log_2(\frac{1}{2})] = 1$
- $H(Y \mid X_2 = normal) = 0$

$\implies H(Y \mid X_3) = [\frac{4}{8} * 1.0 + \frac{4}{8} * 0] = 0.5$
$\implies I(Y; X_3) \approx 0.811 - 0.5 = 0.311$

- Split on X_1 at the root node Since splitting on attribute X_1 gives the highest mutual information, the root node is X_1.

(b)

From the above part, as we can see that the sub-datasets $\mathcal{D}_{(X_1=rain)}$ and $\mathcal{D}_{(X_1=overcast)}$ are pure, there will be no further splitting on those and we will place a leaf node with label assignment decided by majority vote classifier. So, we need to split only on the sub-dataset $\mathcal{D}_{(X_1=sunny)}$. Now, we will use only $\mathcal{D}_{(X_1=sunny)}$ to estimate the probabilities for the next split.

$H(Y) = -\frac{1}{3} * \log_2(\frac{1}{3}) - \frac{2}{3} * \log_2(\frac{2}{3}) \approx 0.918$

For attribute X_2,

- $H(Y \mid X_2 = hot) = 0$
- $H(Y \mid X_2 = cool) = 0$
- $H(Y \mid X_2 = mild) = -[\frac{1}{2} * \log_2(\frac{1}{2}) + \frac{1}{2} * \log_2(\frac{1}{2})] = 1$

$\implies H(Y \mid X_2) = [\frac{2}{3} * 1.0 + \frac{1}{3} * 0] \approx 0.67$
$\implies I(Y; X_2) \approx 0.918 - 0.67 \approx 0.25$

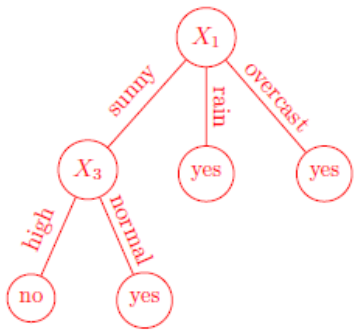
For attribute X_3,

- $H(Y \mid X_3 = high) = 0$
- $H(Y \mid X_3 = normal) = 0$

$\implies H(Y \mid X_3) = [\frac{2}{3} * 0 + \frac{1}{3} * 0] = 0$
$\implies I(Y; X_3) \approx 0.918$

We split using attribute X_3 as it gives the highest mutual information.

(c)



4.

(a) False. SVM gives a solution in both linearly separable data and non-linearly separable data.
(b) False. Logistic regression is a binary classification method. It can also be used for regression.
(c) False. Stochastic gradient descent does not ensure error decrease at each update.
(d) True. We should use validation set not test set to help adjust the hyperparameters.
(e) False. Cross validation is used when data is limited. It has nothing to do with overfitting.